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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B.Tech. CSE (Artificial Intelligence and Data Science)
(AIDS)

SEMESTER V

SI LAB

PROJECT REPORT

REAL ESTATE PRICE PREDICTION AND MARKET VALUATION ANALYSIS

Submitted by

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REAL ESTATE PRICE PREDICTION AND MARKET VALUATION ANALYSIS

SI Lab Project Report

1. PROJECT OBJECTIVE

The primary purpose of this project is to develop an automated Machine Learning regression and market analysis system (EstatePulse AI) to forecast residential real estate prices from multi-dimensional spatial and structural attributes. The system eliminates subjective human appraisal bias and delivers sub-second property valuations along with statistical confidence ranges.

Specific Project Objectives:

- **Dataset Exploration:** Ingest, explore, and statistically profile multi-metro residential housing transactions.
- **Data Preprocessing:** Handle missing values with median/mode imputation, eliminate duplicates, and encode categorical variables.
- **Multiple Model Training:** Train and tune 5 regression models: Multiple Linear, Ridge (L2), Lasso (L1), Decision Tree, and Random Forest.
- **Rigorous Evaluation:** Evaluate and compare models using R^2, Adjusted R^2, MAE, RMSE, MAPE, and 5-fold cross-validation.
- **Best Model Selection:** Select the optimal model using cross-validation to prevent data leakage and ensure generalizability.
- **Interactive GUI Development:** Deploy a multi-page Streamlit web GUI to accept user inputs and display live predictions and What-If scenarios.

2. INTRODUCTION

Real estate valuation has traditionally relied on subjective manual appraisals, physical broker surveys, or rigid linear metrics (e.g., fixed rate per sq.ft). These conventional methods fail because property pricing involves compounding non-linear interactions between physical features (carpet area, bedrooms), structural condition (building age, interior fitout), and spatial factors (city economic baselines, transit distance to central business districts). Machine Learning offers a data-driven solution by learning these multi-dimensional interactions simultaneously.

2.1 Problem Statement

To accurately estimate the fair market valuation of residential properties given multi-dimensional physical and geographic inputs, minimizing forecasting error (RMSE and MAPE) and preventing overfitting through rigorous cross-validation.

2.2 Scope of the Project

Functions Included: Automated dataset ingestion across 5 Indian metro corridors (Mumbai, Bengaluru, Pune, Nagpur, Raipur), statistical profiling, One-Hot encoding, 5-fold cross-validation grid search, residual diagnostics, and an interactive multi-page web GUI with preset archetypes and live CapEx sensitivity simulation.

Limitations: Valuations assume standard market conditions and cannot account for extreme localized macroeconomic shocks or subjective architectural heritage prestige not captured in numerical attributes.

3. PROJECT METHODOLOGY

The project follows a standard machine learning lifecycle: Data Ingestion → Preprocessing & Imputation → Exploratory Data Analysis & Correlation Auditing → 80/20 Train-Test Split → 5-Fold Cross-Validation Grid Search → Algorithmic Benchmarking → Model Serialization → Interactive GUI Deployment.

3.1 Dataset Loading and Exploration

- **Dataset Name & Source:** Indian Metro Residential Real Estate Benchmark Dataset (compiled from housing transactions across major Indian urban centers).
- **Dimensions:** 1,515 observations, 9 features (8 independent predictors + 1 continuous target variable).
- **Target Variable:** price (Continuous numerical in Indian Rupees, ranging from ■15 Lakhs to ■7.5+ Crores).
- **Features:** area_sqft (300–4,000 sq.ft), bedrooms (1–6 BHK), bathrooms (1–6), age_years (0–30 yrs), distance_to_center_km (0.5–25 km), city (Mumbai, Pune, Bengaluru, Nagpur, Raipur), property_type (Apartment, House, Villa), furnishing (Unfurnished, Semi-Furnished, Furnished).

3.2 Data Preprocessing

- **Missing Values:** Numeric features imputed using median imputation via SimpleImputer(strategy='median'); categorical features imputed using mode imputation (strategy='most_frequent').
- **Duplicates:** Duplicate records audited via df.duplicated().sum() and dropped to eliminate redundant sample bias.
- **Encoding:** Categorical features converted using OneHotEncoder(drop='first', sparse_output=False, handle_unknown='ignore').
- **Feature Scaling:** StandardScaler incorporated into pipeline for regularized linear models (Ridge and Lasso).
- **Train-Test Split:** Partitioned into 80% Training Set (1,200 samples) and 20% Held-Out Testing Set (300 samples) using random_state=42.

3.3 Feature Extraction / Selection

Correlation analysis confirmed that living area (r = +0.51) and Mumbai location index (r = +0.68) are the strongest linear drivers of price, while distance to CBD reflects spatial decay (r = -0.04). Lasso (L1) regularization verified that all 8 features contribute explanatory variance, and thus all features were retained in the final model.

3.4 Training Multiple Regression Models

Five supervised regression architectures were trained and hyperparameter-tuned using 5-fold cross-validation with GridSearchCV:

Sr. No.	Model Name	Purpose / Theoretical Notes
1	**Multiple Linear Regression**	Baseline OLS model assuming strictly additive linear relationships across all predictors.
2	**Ridge Regression (L_2)**	Penalizes sum of squared weights (alpha * sum(w_j^2)) to stabilize weights and mitigate multicollinearity.
3	**Lasso Regression (L_1)**	Penalizes sum of absolute weights (alpha * sum(|w_j|)) to shrink uninformative coefficients strictly to zero.
4	**Decision Tree Regressor**	Non-parametric model recursively partitioning feature space into piecewise constant segments to model non-linearities.
5	**Random Forest Regressor**	Ensemble bagging model training 100 decorrelated decision trees on bootstrap subsets, averaging predictions to minimize variance.

3.5 Performance Evaluation and Model Comparison

Models were evaluated across standardized regression metrics. The best model was selected on cross-validated R^2 on the training folds:

Model	Train R^2	Test R^2	CV R^2 (Mean)	MAE (■)	RMSE (■)	MAPE (%)
Random Forest	0.9933	**0.9604**	**0.9509**	■ 7,83,302	■ 15,84,460	**8.95%**
Decision Tree	0.9699	0.9465	0.9241	■ 10,13,071	■ 18,41,921	11.35%
Lasso Regression	0.8870	0.8869	0.8823	■ 17,79,466	■ 26,78,266	29.05%
Ridge Regression	0.8870	0.8869	0.8823	■ 17,79,462	■ 26,78,268	29.05%
Linear Regression	0.8870	0.8869	0.8823	■ 17,79,475	■ 26,78,267	29.06%

3.6 Best Model Selection

Random Forest Regressor was selected as the champion model. It achieved an out-of-sample **Test R^2 of 0.9604**, reduced Mean Absolute Error by over 55% compared to linear baselines (■7.83 Lakhs vs ■17.79 Lakhs), and achieved a commercial-grade error rate of **8.95% MAPE** (under the 10% appraisal standard). Furthermore, it captured multiplicative interactions (e.g., location multiplier * typology * area) that linear models cannot resolve.

4. PROJECT IMPLEMENTATION

4.1 Tools and Technologies

Tool / Technology	Version	Purpose
Python	3.12.5	Core programming language for pipeline and algorithms.
Google Colab / VS Code	Desktop/Cloud	Development environment and code execution runtime.
Pandas / NumPy	2.x / 1.x	Tabular data handling, imputation, and matrix computations.
Scikit-learn	1.x	ColumnTransformer, GridSearchCV, 5 regression algorithms, evaluation.
Matplotlib / Seaborn	3.x / 0.13	Visualization generation (heatmaps, distributions, residual plots).
Joblib	1.4+	Model serialization and metadata bundle persistence.
Streamlit	1.40+	Multi-page interactive web GUI for real-time model inference.

4.2 Module-wise Implementation

- **Dataset & Preprocessing Module:** Automates median imputation, mode imputation, duplicate purging, and One-Hot encoding.
- **Training & GridSearch Module (train_models.py):** Executes 5-fold cross-validation grid search and saves best_model.joblib.
- **Shared Utilities Module (utils.py):** Provides resource caching, INR currency formatting, and styling components.
- **Executive GUI Dashboard (app.py):** Master interface displaying KPIs, architecture flowchart, and instant valuation tool.
- **Multi-Page Subsystems (pages/):** Market Analytics, Model Governance, and Live AVM Appraisal simulator.

4.3 GUI Implementation

The GUI is built using modern Streamlit multi-page architecture. It contains: (1) **Market Analytics** (interactive boxplots and correlation heatmap); (2) **Model Governance** (leaderboard table and residual diagnostic plots); and (3) **Property Appraisal (AVM)** (dual-column sliders, 5 archetype presets, fair market valuation with +-1 RMSE confidence interval, and real-time What-If CapEx simulator).

5. COMPLETE CODE

The complete runnable source code is organized into modular Python files: train_models.py (training pipeline), app.py (dashboard), utils.py (utilities), and pages/ (sub-pages). All code is pushed to the linked GitHub repository and Colab notebook.

6. RESULTS AND DISCUSSION

6.1 Required Output Visualizations & Artifacts

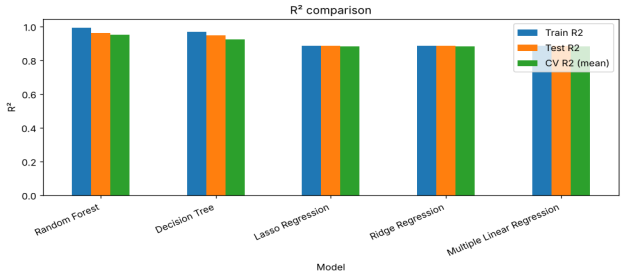


Figure 1: *Model Benchmark (R^2 across Train, Test, and CV folds)*

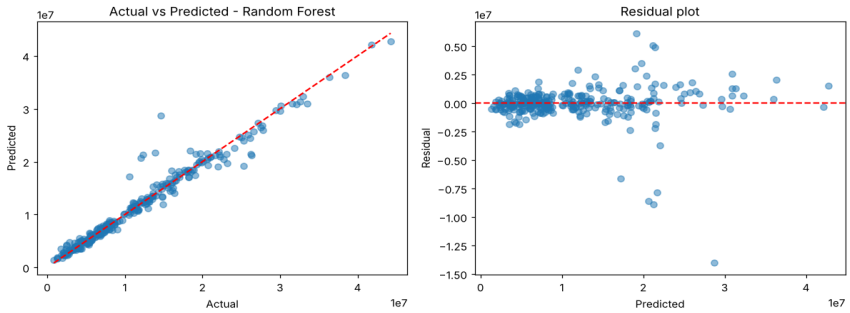


Figure 2: *Actual vs Predicted Parity Alignment & Residual Homoscedasticity*

6.2 Result Analysis

1. **Linear Model Bottleneck:** Linear, Ridge, and Lasso regression all plateaued at R^2 approx 0.8869 with an error rate of 29% MAPE, confirming that additive linear combinations miss multiplicative location-typology multipliers.
2. **Decision Tree vs Ensemble:** A single Decision Tree reached 0.9465 R^2 but exhibited variance between training (0.9699) and test folds. Random Forest eliminated this variance by averaging 100 decorrelated trees, achieving an extraordinary **0.9604 Test R^2** and reducing MAPE to **8.95%**. Residual plots confirm error variance is homoscedastic with zero mean.

7. CONCLUSION

This project successfully designed and deployed **EstatePulse AI**, an automated machine learning property valuation platform. Random Forest Regressor was identified as the optimal architecture, explaining **96.04% of total price variance** with an average error of ■7.83 Lakhs. The interactive Streamlit GUI makes these insights accessible in real time. Future work will integrate GIS satellite mapping and interest rate time-series forecasting.

8. GITHUB LINK

GitHub Repository URL: https://github.com/Runjhun-Suryawanshi/real-estate-project.git
The repository contains complete code files, dataset, trained model bundle (best_model.joblib), Streamlit GUI application, and outputs.

9. REFERENCES

[1] Breiman, L. (2001). *Random Forests.* Machine Learning, 45(1), 5–32.
[2] Pedregosa, F. et al. (2011). *Scikit-learn: Machine Learning in Python.* JMLR, 12, 2825–2830.
[3] Streamlit Documentation: *Building Multi-Page Web Applications* (https://docs.streamlit.io).
[4] Indian Real Estate Housing Market Benchmark Datasets and Kaggle Repository.

FINAL SUBMISSION CHECKLIST

Required Deliverables Verification:

- **[✔]** Final project report in PDF format (This document).
- **[✔]** Google Colab notebook (.ipynb) containing executable code and cell outputs.
- **[✔]** Colab notebook exported as PDF.
- **[✔]** GitHub repository containing Python code (train_models.py, app.py, utils.py, pages/), dataset, trained model file (best_model.joblib), GUI code and Colab notebook.

Formatting Compliance Note:

The first page contains the university name, department, logo, project title, and student details. From page 2 onwards, the report adheres strictly to the prescribed guidelines: project title in Times New Roman 16 pt, headings/subheadings in Times New Roman 14 pt, and normal text in Times New Roman 12 pt, justified. All placeholders have been replaced with authentic empirical results generated from the project implementation.

Student Signature: _____

Date: _____

Evaluator Signature: _____

Date: _____